

[illegible]

a)  $t = ?$ ,  $\lambda_{\text{eff}}(s) = P_0$   
 $P(t) = 3P_0$   
 $\lambda(s) = \frac{P(s)}{P_0} = 3$   
 $\lambda(s) = \frac{R_0(s)}{R_0(0)} = 3$   
 $\lambda(s) = e^{\frac{R_0(s)}{R_0(0)} - 1} = 3$   
 $\ln(3) = \frac{R_0(s)}{R_0(0)} - 1$   
 $\ln(3) + 1 = \frac{R_0(s)}{R_0(0)}$   
 $\ln(3) + 1 = \frac{R_0(0) + \lambda R_0(0) t}{R_0(0)}$   
 $\ln(3) + 1 = 1 + \lambda t$   
 $\ln(3) = \lambda t$   
 $t = \frac{\ln(3)}{\lambda} = \frac{\ln(3)}{4} \approx 0.347 \text{ a.u.}$

b)  $t = ?$ ,  $\lambda_{\text{eff}}(s) = P_0$   
 $P(t) = 4P_0$   
 $\lambda(s) = \frac{P(s)}{P_0} = 4$   
 $\lambda(s) = \frac{R_0(s)}{R_0(0)} = 4$   
 $\lambda(s) = e^{\frac{R_0(s)}{R_0(0)} - 1} = 4$   
 $\ln(4) = \frac{R_0(s)}{R_0(0)} - 1$   
 $\ln(4) + 1 = \frac{R_0(s)}{R_0(0)}$   
 $\ln(4) + 1 = \frac{R_0(0) + \lambda R_0(0) t}{R_0(0)}$   
 $\ln(4) + 1 = 1 + \lambda t$   
 $\ln(4) = \lambda t$   
 $t = \frac{\ln(4)}{\lambda} = \frac{\ln(4)}{4} \approx 0.347 \text{ a.u.}$